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Errata

Doc No. : Er-Vx86-140423

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Subject:

RTDSC instruction will get wrong time-stamp value.

Description:

Using RDTSC get time-stamp value repeatedly, the 64-bit time-stamp value has possibility to be wrong in low 32 bit part.).

Affected SoCs:

Vortex86DX, Vortex86MX+,Vortex86EX, Vortex86DX2

Root Cause:

CPU time-stamp counter is increased with every CPU clock tick, RDTSC instruction behave that update time-stamp value to CPU general register EDX:EAX. The update behavior is multi-cycle and update EAX firstly then EDX. So EAX value is CPU time-stamp tick N and EDX value is CPU time-stamp tick N+1. When time-stamp value is near overflow in low 32bits part, this get time-stamp value behavior will get the wrong value.

Work Around :

When using RDTSC instruction, check time-stamp value bit31 to bit 4. If time stamp value is XXXXXXXX_FFFFFFFX (X means don't care), skip this result and get time-stamp value again.
Below is RDTSC software solution example code (Assembly code):

_ _ _ _ _

```
re_read:
        RDTSC
        cmp eax, 0xfffffff0
        ja re_read
```

_ _ _ _ _

END